HENSELIAN 对的五种等价判据

Counting scope. The primary problem is Tag 09XI, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

Copyright (C) 2005–2025 Johan de Jong. Stacks Project authors. Source: Tag 09XI.

Difficulty H2: 本题将 Henselian 因式分解、étale 截面提升、有限或整代数的幂等元提升及 Gabber 特殊多项式条件相互转化。证明用幂等元分解与特征多项式恢复因式分解，并经整闭包和有限子代数把特殊多项式根提升转成 étale 截面。即使给定辅助有限性引理，双向结构转换仍超出定义核查，判 H2。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, more-algebra.tex, lines 2552–2734. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Definition 0.1. A *henselian pair* is a pair (A, I) satisfying

(1) I is contained in the Jacobson radical of A, and
(2) for any monic polynomial $f \in A[T]$ and factorization $\overline{f} = g_0 h_0$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$, there exists a factorization $f = gh$ in $A[T]$ with g, h monic and $g_0 = \overline{g}$ and $h_0 = \overline{h}$.

Lemma 0.2. *Let (A, I) be a Zariski pair. Let $A \to B$ be a finite ring map. Assume*

(1) *$B/IB = B_1 \times B_2$ is a product of A/I-algebras*
(2) *$A/I \to B_1/IB_1$ is surjective,*
(3) *$b \in B$ maps to $(1, 0)$ in the product.*

Then there exists a monic $f \in A[x]$ with $f(b) = 0$ and $f \bmod I = (x-1)x^d$ for some $d \geq 1$.

证明. By Lemma lemma lift idempotent upstairs (Tag 07M4) we can find an étale ring map $A \to A'$ inducing an isomorphism $A/I \to A'/IA'$ such that $B' = B \otimes_A A'$ contains an idempotent e' lifting the image of b in B'/IB'. Consider the corresponding A'-algebra decomposition

$$B' = B'_1 \times B'_2$$

which is compatible with the one given in the lemma upon reduction modulo I. The map $A' \to B'_1$ is surjective modulo IA'. By Nakayama's lemma (Algebra, Lemma lemma NAK (Tag 00DV)) we can find $i \in IA'$ such that after replacing A' by A'_{1+i} the map $A' \to B'_1$ is surjective. Observe that the image $b'_1 \in B'_1$ of b satisfies $b'_1 - 1 \in IB'_1$. Thus we may pick $a' \in IA'$ mapping to $b'_1 - 1$. On the other hand, the image $b'_2 \in B'_2$ of b is in IB'_2. By Algebra, Lemma lemma integral integral over ideal (Tag 00H5) there exist a monic polynomial $g(x) = x^d + \sum a'_j x^j$ of degree d with $a'_j \in IA'$ such that $g(b'_2) = 0$ in B'_2. Thus the image $b' = (b'_1, b'_2) \in B'$ of b is a root of the polynomial $(x - 1 - a')g(x)$. We conclude that

$$(b'-1)(b')^d \in \sum\nolimits_{j=0,\ldots,d} IA' \cdot (b')^j$$

We claim that this implies

$$(b-1)b^d \in \sum\nolimits_{j=0,\ldots,d} I \cdot b^j$$

in B. For this it is enough to see that the ring map $A \to A'$ is faithfully flat, because the condition is that the image of $(b-1)b^d$ is zero in $B/\sum_{j=0,\ldots,d} Ib^j$ (use Algebra, Lemma lemma faithfully flat universally injective (Tag 05CK)). The map $A \to A'$ flat because it is étale (Algebra, Lemma lemma etale (Tag 00U2)). On the other hand, the induced map on spectra is open (see Algebra, Proposition proposition fppf open (Tag 00I1) and use previous lemma referenced) and the image contains $V(I)$. Since I is contained in the Jacobson radical of A we conclude. □

Lemma 0.3. *Let (A, I) be a pair. Let $A \to B$ be a finite type ring map such that $B/IB = C_1 \times C_2$ with $A/I \to C_1$ finite. Let B' be the integral closure of A in B. Then we can write $B'/IB' = C_1 \times C_2'$ such that the map $B'/IB' \to B/IB$ preserves product decompositions and there exists a $g \in B'$ mapping to $(1, 0)$ in $C_1 \times C_2'$ with $B'_g \to B_g$ an isomorphism.*

证明. Observe that $A \to B$ is quasi-finite at every prime of the closed subset $T = \mathrm{Spec}(C_1) \subset \mathrm{Spec}(B)$ (this follows by looking at fibre rings, see Algebra, Definition 0.5). Consider the diagram of topological spaces

$$\begin{array}{ccc} \mathrm{Spec}(B) & \xrightarrow{\quad\phi\quad} & \mathrm{Spec}(B') \\ & {\scriptstyle\psi}\searrow \quad \swarrow{\scriptstyle\psi'} & \\ & \mathrm{Spec}(A) & \end{array}$$

By Algebra, Theorem theorem main theorem (Tag 00Q9) for every $\mathfrak{p} \in T$ there is a $h_\mathfrak{p} \in B'$, $h_\mathfrak{p} \notin \mathfrak{p}$ such that $B'_h \to B_h$ is an isomorphism. The union $U = \bigcup D(h_\mathfrak{p})$ gives an open $U \subset \mathrm{Spec}(B')$ such that $\phi^{-1}(U) \to U$ is a homeomorphism and $T \subset \phi^{-1}(U)$. Since T is open in $\psi^{-1}(V(I))$ we conclude that $\phi(T)$ is open in $U \cap (\psi')^{-1}(V(I))$. Thus $\phi(T)$ is open in $(\psi')^{-1}(V(I))$. On the other hand, since C_1 is finite over A/I it is finite over B'. Hence $\phi(T)$ is a closed subset of $\mathrm{Spec}(B')$ by Algebra, Lemmas lemma going up closed (Tag 00HZ) and lemma integral going up (Tag 00GU). We conclude that $\mathrm{Spec}(B'/IB') \supset \phi(T)$ is open and closed. By Algebra, Lemma lemma disjoint implies product (Tag 00EM) we get a corresponding product decomposition $B'/IB' = C_1' \times C_2'$. The map $B'/IB' \to B/IB$ maps C_1' into C_1 and C_2' into C_2 as one sees by looking at what happens on spectra (hint: the inverse image of $\phi(T)$ is exactly T; some details omitted). Pick a $g \in B'$ mapping to $(1, 0)$ in $C_1' \times C_2'$ such that $D(g) \subset U$; this is possible because $\mathrm{Spec}(C_1')$ and $\mathrm{Spec}(C_2')$ are disjoint and closed in $\mathrm{Spec}(B')$ and $\mathrm{Spec}(C_1')$ is contained in U. Then $B'_g \to B_g$ defines a homeomorphism on spectra and an isomorphism on local rings (by our choice of U above). Hence it is an isomorphism, as follows for example from Algebra, Lemma lemma characterize zero local (Tag 00HN). Finally, it follows that $C_1' = C_1$ and the proof is complete. □

Lemma 0.4. *Let (A, I) be a Zariski pair. Let $A \to B$ be a flat, integral, finitely presented ring map such that $A/I \to B/IB$ is an isomorphism. Then $A \to B$ is an isomorphism.*

Definition 0.5. Let $R \to S$ be a finite type ring map. Let $\mathfrak{q} \subset S$ be a prime.

(1) If the equivalent conditions of Lemma lemma isolated point fibre (Tag 00PK) are satisfied then we say $R \to S$ is *quasi-finite at* $\mathfrak{q}$.
(2) We say a ring map $A \to B$ is *quasi-finite* if it is of finite type and quasi-finite at all primes of B.

Primary problem: Tag 09XI

Lemma 0.6. *Let (A, I) be a pair. The following are equivalent*

(1) (A, I) *is a henselian pair,*
(2) *given an étale ring map* $A \to A'$ *and an* A*-algebra map* $\sigma : A' \to A/I$*, there exists an* A*-algebra map* $A' \to A$ *lifting* σ*,*
(3) *for any finite* A*-algebra* B *the map* $B \to B/IB$ *induces a bijection on idempotents,*
(4) *for any integral* A*-algebra* B *the map* $B \to B/IB$ *induces a bijection on idempotents, and*
(5) *(Gabber)* I *is contained in the Jacobson radical of* A *and every monic polynomial* $f(T) \in A[T]$ *of the form*

$$f(T) = T^n(T-1) + a_nT^n + \ldots + a_1T + a_0$$

with $a_n, \ldots, a_0 \in I$ *and* $n \geq 1$ *has a root* $\alpha \in 1 + I$*.*

Moreover, in part (5) the root is unique.

证明. Assume (2) holds. Then I is contained in the Jacobson radical of A, since otherwise there would be a nonunit $f \in A$ congruent to 1 modulo I and the map $A \to A_f$ would contradict (2). Hence $IB \subset B$ is contained in the Jacobson radical of B for B integral over A because $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is closed by Algebra, Lemmas lemma going up closed (Tag 00HZ) and lemma integral going up (Tag 00GU). Thus the map from idempotents of B to idempotents of B/IB is injective by Lemma lemma idempotents determined modulo radical (Tag 09XF). On the other hand, since (2) holds, every idempotent of B/IB lifts to an idempotent of B by Lemma lemma lift idempotent upstairs (Tag 07M4). In this way we see that (2) implies (4).

The implication (4) $\Rightarrow$ (3) is trivial.

Assume (3). Let $\mathfrak{m}$ be a maximal ideal and consider the finite map $A \to B = A/(I \cap \mathfrak{m})$. The condition that $B \to B/IB$ induces a bijection on idempotents implies that $I \subset \mathfrak{m}$ (if not, then $B = A/I \times A/\mathfrak{m}$ and $B/IB = A/I$). Thus we see that I is contained in the Jacobson radical of A. Let $f \in A[T]$ be monic and suppose given a factorization $\overline{f} = g_0h_0$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$. Set $B = A[T]/(f)$. Let $\overline{e}$ be the idempotent of B/IB corresponding to the decomposition

$$B/IB = A/I[T]/(g_0) \times A/I[T]/(h_0)$$

of A-algebras. Let $e \in B$ be an idempotent lifting $\overline{e}$ which exists as we assumed (3). This gives a product decomposition

$$B = eB \times (1-e)B$$

Note that B is free of rank $\deg(f)$ as an A-module. Hence eB and $(1-e)B$ are finite locally free A-modules. However, since eB and $(1-e)B$ have constant rank $\deg(g_0)$ and $\deg(h_0)$ over A/I we find that the same is true over $\mathrm{Spec}(A)$. We conclude that

$$\begin{aligned} f &= \mathrm{CharPol}_A(T : B \to B) \\ &= \mathrm{CharPol}_A(T : eB \to eB)\mathrm{CharPol}_A(T : (1-e)B \to (1-e)B) \end{aligned}$$

is a factorization into monic polynomials reducing to the given factorization modulo I. Here $\mathrm{CharPol}_A$ denotes the characteristic polynomial of an endomorphism of a finite locally free module over A. If the module is free the $\mathrm{CharPol}_A$ is defined as the characteristic polynomial of the corresponding matrix and in general one uses Algebra, Lemma lemma standard covering (Tag 00EJ) to glue. Details omitted. Thus (3) implies (1).

Assume (1). Let f be as in (5). The factorization of f mod I as T^n times $T-1$ lifts to a factorization $f = gh$ with g and h monic by Definition 0.1. Then h has to have degree 1 and we see that f has a root reducing to 1 modulo I. Finally, I is contained in the Jacobson radical by the definition of a henselian pair. Thus (1) implies (5).

Before we give the proof of the last step, let us show that the root α in (5), if it exists, is unique. Namely, due to the explicit shape of $f(T)$, we have $f'(\alpha) \in 1 + I$ where f' is the derivative of f with respect to T. An elementary argument shows that

$$f(T) = f(\alpha + T - \alpha) = f(\alpha) + f'(\alpha) \cdot (T - \alpha) \bmod (T - \alpha)^2 A[T]$$

This shows that any other root $\alpha' \in 1 + I$ of $f(T)$ satisfies $0 = f(\alpha') - f(\alpha) = (\alpha' - \alpha)(1 + i)$ for some $i \in I$, so that, since $1 + i$ is a unit in A, we have $\alpha = \alpha'$.

Assume (5). We will show that (2) holds, in other words, that for every étale map $A \to A'$, every section $\sigma : A' \to A/I$ modulo I lifts to a section $A' \to A$. Since $A \to A'$ is étale, the section σ determines a decomposition

$$A'/IA' \cong A/I \times C \tag{0.6.1}$$

of A/I-algebras. Namely, the surjective ring map $A'/IA' \to A/I$ is étale by Algebra, Lemma lemma map between etale (Tag 00U7) and then we get the desired idempotent by Algebra, Lemma lemma surjective flat finitely presented (Tag 00U8). We will show that this decomposition lifts to a decomposition

$$A' \cong A'_1 \times A'_2 \tag{0.6.2}$$

of A-algebras with A'_1 integral over A. Then $A \to A'_1$ is integral and étale and $A/I \to A'_1/IA'_1$ is an isomorphism, thus $A \to A'_1$ is an isomorphism by Lemma 0.4 (here we also use that an étale ring map is flat and of finite presentation, see Algebra, Lemma lemma etale (Tag 00U2)).

Let B' be the integral closure of A in A'. By Lemma 0.3 we may decompose

$$B'/IB' \cong A/I \times C' \tag{0.6.3}$$

as A/I-algebras compatibly with (0.6.1) and we may find $b \in B'$ that lifts $(1, 0)$ such that $B'_b \to A'_b$ is an isomorphism. If the decomposition (0.6.3) lifts to a decomposition

$$B' \cong B'_1 \times B'_2 \tag{0.6.4}$$

of A-algebras, then the induced decomposition $A' = A'_1 \times A'_2$ will give the desired (0.6.2): indeed, since b is a unit in B'_1 (details omitted), we will have $B'_1 \cong A'_1$, so that A'_1 will be integral over A.

Choose a finite A-subalgebra $B'' \subset B'$ containing b (observe that any finitely generated A-subalgebra of B' is finite over A). After enlarging B'' we may assume b maps to an idempotent in B''/IB'' producing

$$B''/IB'' \cong C''_1 \times C''_2 \tag{0.6.5}$$

Since $B'_b \cong A'_b$ we see that B'_b is of finite type over A. Say B'_b is generated by $b_1/b^n, \ldots, b_t/b^n$ over A and enlarge B'' so that $b_1, \ldots, b_t \in B''$. Then $B''_b \to B'_b$ is surjective as well as injective, hence

an isomorphism. In particular, we see that $C_1'' = A/I$! Therefore $A/I \to C_1''$ is an isomorphism, in particular surjective. By Lemma 0.2 we can find an $f(T) \in A[T]$ of the form

$$f(T) = T^n(T-1) + a_n T^n + \ldots + a_1 T + a_0$$

with $a_n, \ldots, a_0 \in I$ and $n \geq 1$ such that $f(b) = 0$. In particular, we find that B' is a $A[T]/(f)$-algebra. By (5) we deduce there is a root $a \in 1 + I$ of f. This produces a product decomposition $A[T]/(f) = A[T]/(T-a) \times D$ compatible with the splitting (0.6.3) of B'/IB'. The induced splitting of B' is then a desired (0.6.4). $\square$